

Операционные системы

Командная оболочка Midnight Commander

Харламова Арина Александровна

2026-09-04

Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить возможности Midnight Commander
- 2 Изучить редактор Midnight Commander

2. Процесс выполнения лабораторной работы

Работа с Midnight Commander

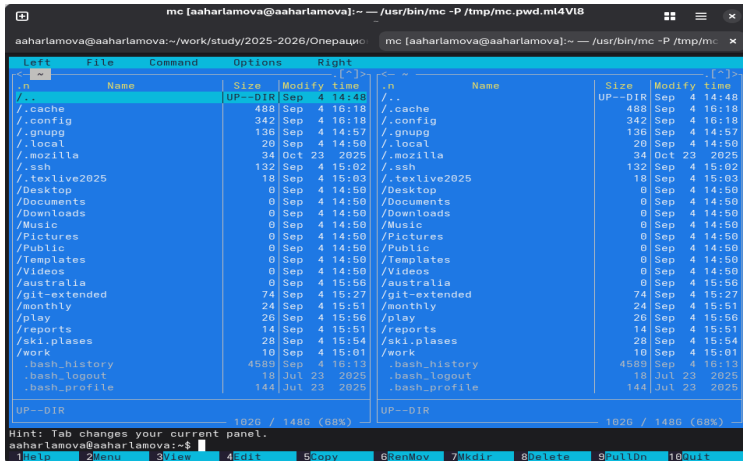


Рисунок 1: Запуск ms

Работа с Midnight Commander

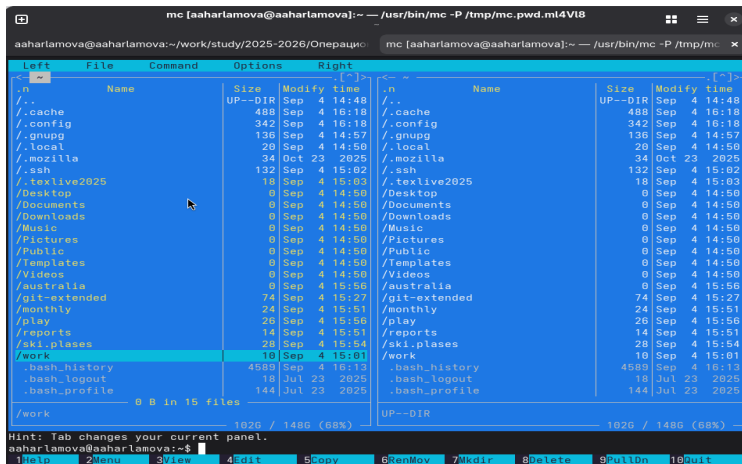


Рисунок 2: Выделение

Работа с Midnight Commander

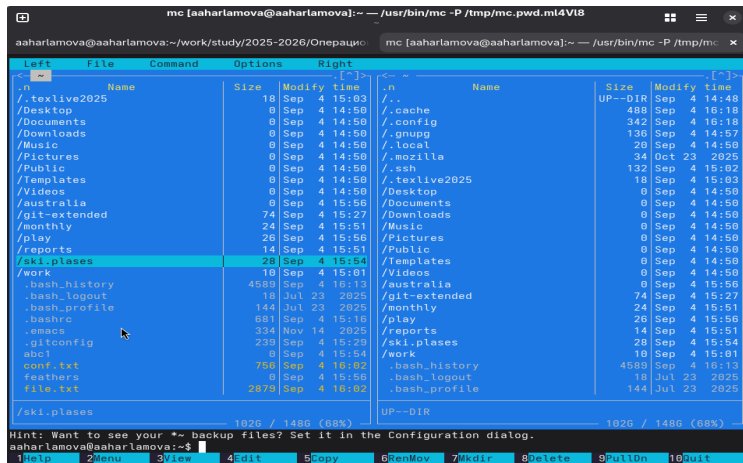


Рисунок 3: Отмена

Работа с Midnight Commander

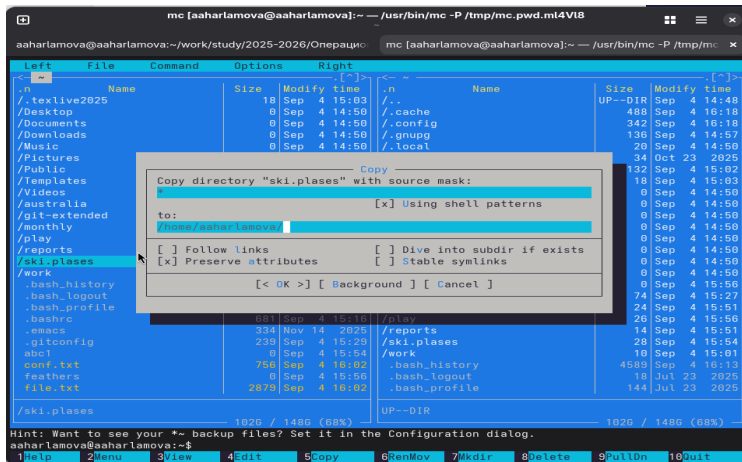


Рисунок 4: Копирование

Работа с Midnight Commander

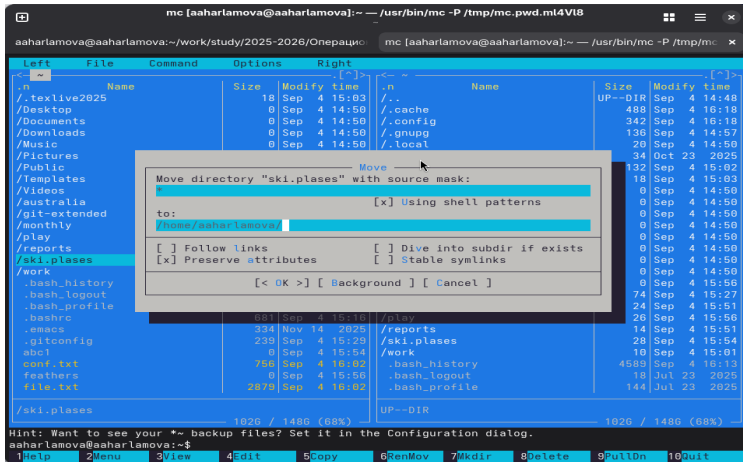


Рисунок 5: Перемещение

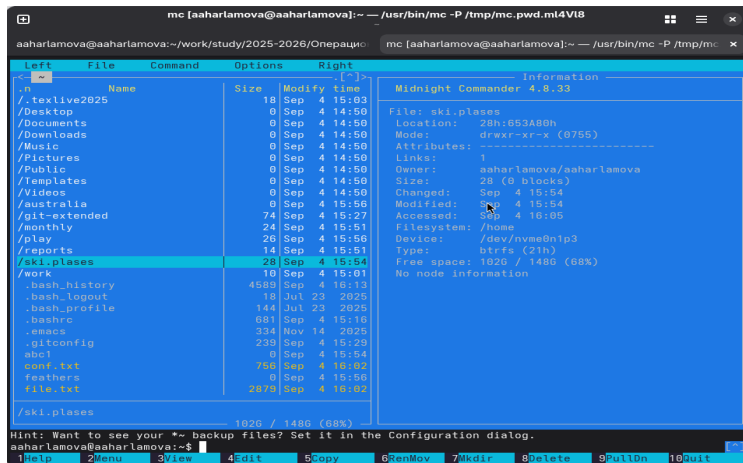


Рисунок 6: Информация

Работа с Midnight Commander

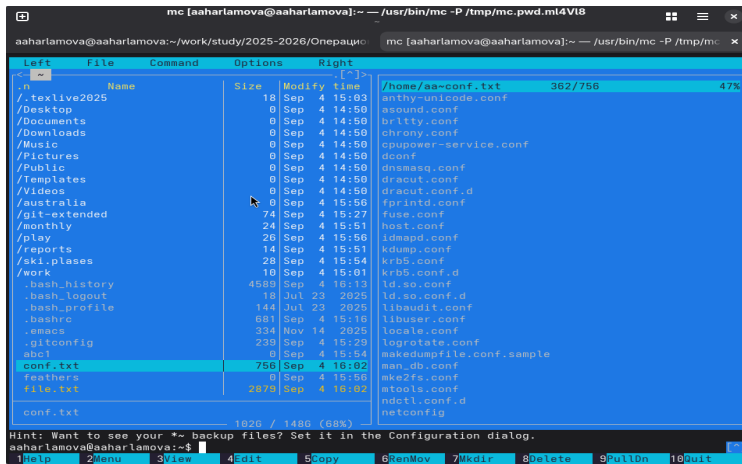


Рисунок 7: Быстрый просмотр

Работа с Midnight Commander

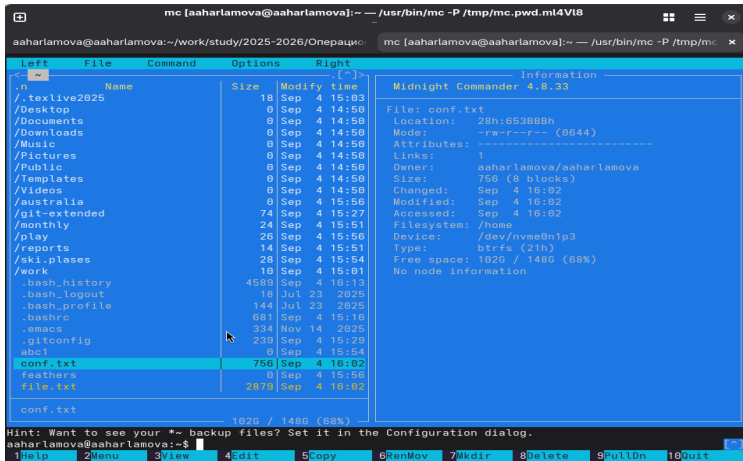


Рисунок 8: Информация

Работа с Midnight Commander

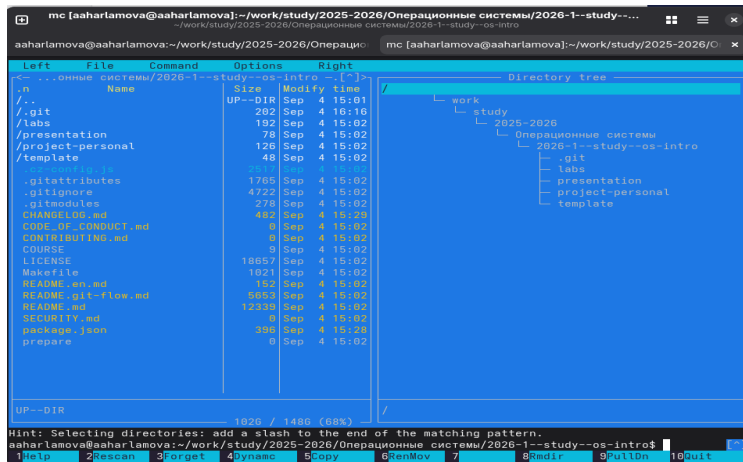
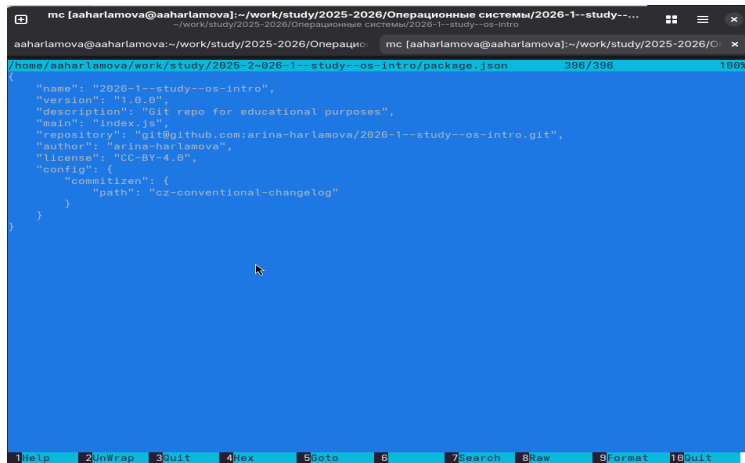


Рисунок 9: Дерево каталогов

Работа с Midnight Commander



The screenshot shows the Midnight Commander (MC) interface. The top bar displays the current directory: `mc [aaharlamova@aaharlamova]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro`. The main pane shows the contents of the file `package.json` located at `/home/aaharlamova/work/study/2025-2026-1--study--os-intro/package.json`. The file size is 396/396 bytes, and it is 100% zoomed. The file content is displayed in a monospaced font on a blue background. The bottom status bar shows various keyboard shortcuts: 1 Help, 2 UnWrap, 3 Quit, 4 Hex, 5 Goto, 6, 7 Search, 8 Raw, 9 Format, 10 Quit.

```
{
  "name": "2026-1--study--os-intro",
  "version": "1.0.0",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:arina-harlamova/2026-1--study--os-intro.git",
  "author": "arina-harlamova",
  "license": "CC-BY-4.0",
  "config": {
    "commitizen": {
      "path": "cz-conventional-changelog"
    }
  }
}
```

Рисунок 10: Просмотр содержимого текстового файла

Работа с Midnight Commander

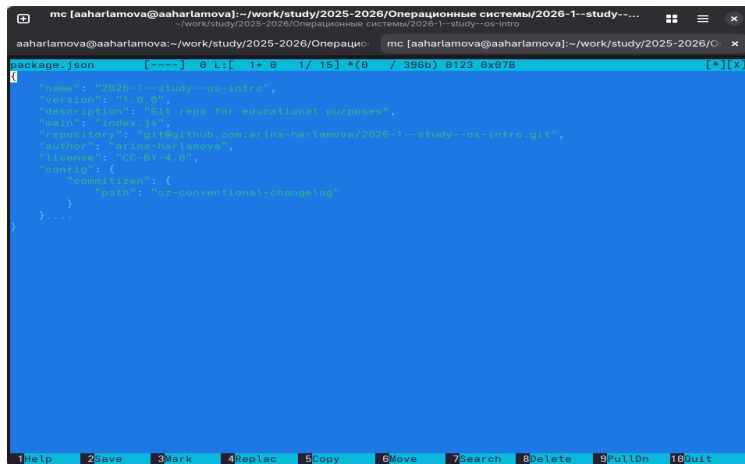


Рисунок 11: Отредактируем содержимое текстового файла без сохранения результатов

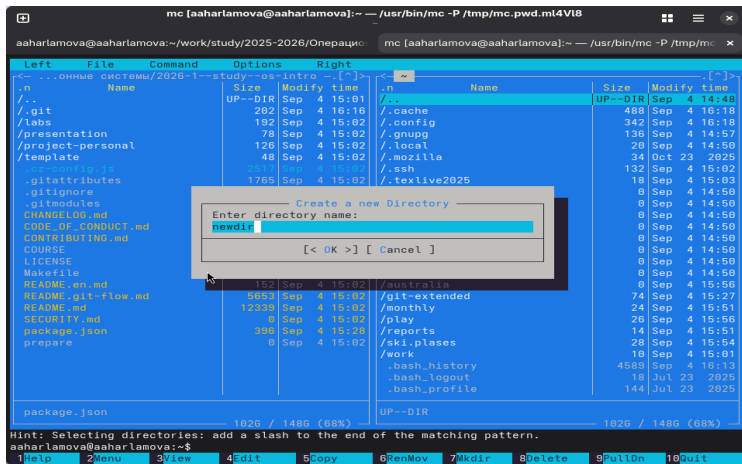


Рисунок 12: Создание каталога

Работа с Midnight Commander

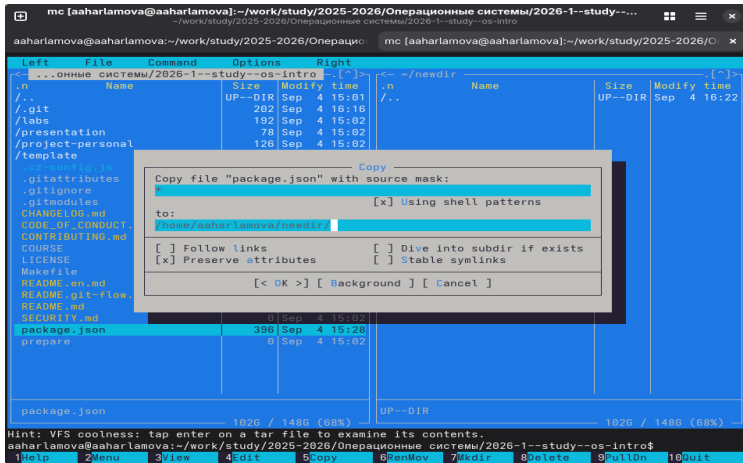


Рисунок 13: Копирование в файлов в созданный каталог

Работа с Midnight Commander

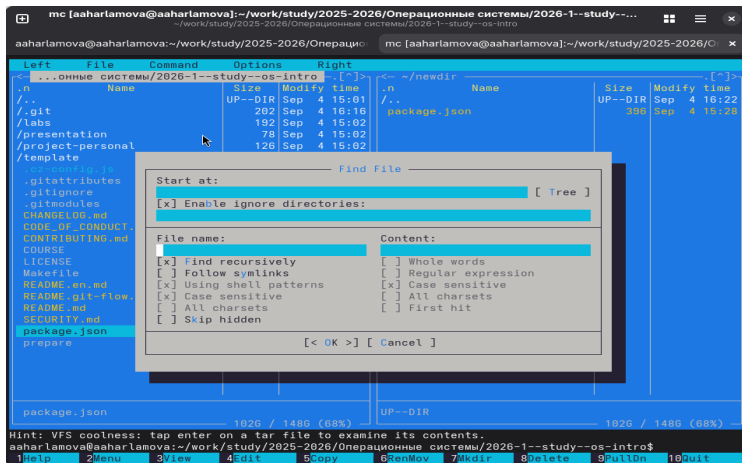


Рисунок 14: Поиск файлов

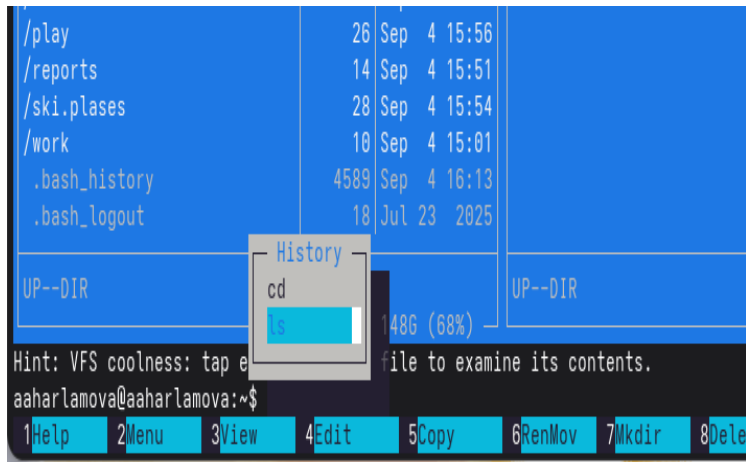


Рисунок 15: История команд

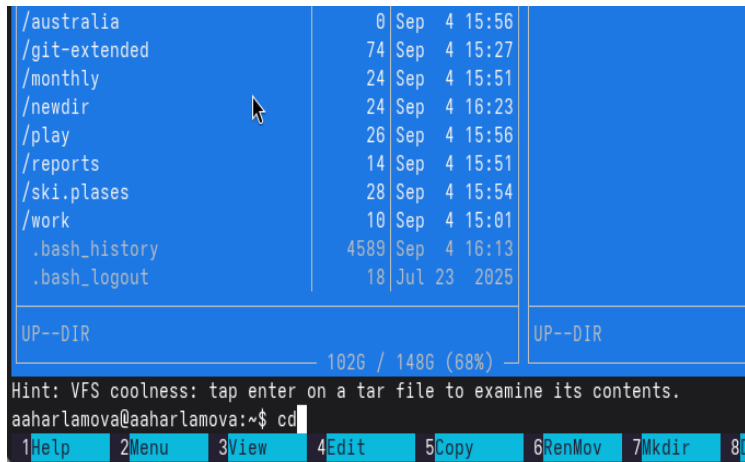
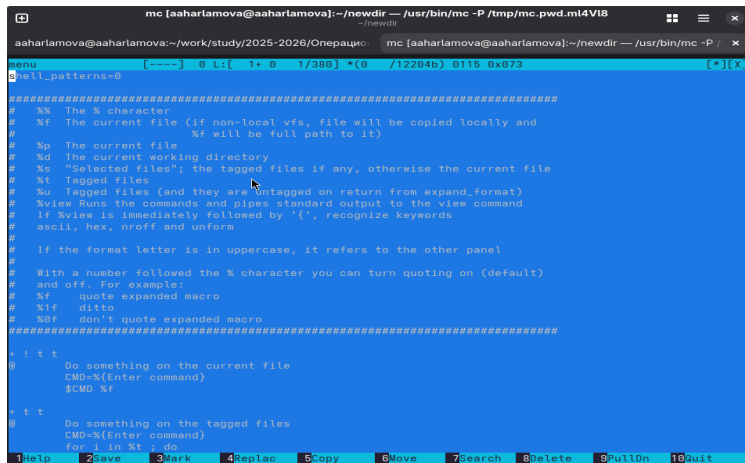


Рисунок 16: Переход в домашний каталог

```
mc [aaharlamova@aaharlamova]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.m14Vl8
aaharlamova@aaharlamova:~/work/study/2025-2026/Операцио: mc [aaharlamova@aaharlamova]:~/newdir — /usr/bin/mc -P /
mc.ext.ini [----] 0 L: [ 1+ 0 1/1183] *(0 /28366b) 0035 0x023 [+](x)
# Midnight Commander 4.0 extension file
#
# Warning: The structure of this file has been completely changed with the version 4.0!
#
# All lines starting with # or empty lines are ignored.
#
# IMPORTANT: mc scans this file only upon first use or after editing it using the
# mc "Edit extension file" command (F9-c-e). If you edit this file in any other way
# while mc is running, you will need to press F9-c-e and exit the editor for your
# changes to take effect, or exit mc and start it again.
#
# Section name can be anything with following exceptions:
#   there are two reserved section names:
#       mc.ext.ini
#       Default
#   special name pattern:
#       Include/xxxxx
# See below for more details.
#
# Section [mc.ext.ini] is mandatory. It contains file metadata.
# "Version" parameter is mandatory. It contains the file format version.
#
# Section [Default] is optional. It is applied only if no other match was found.
#
# Sections like [Include/xxxx] can be referenced as "Include=xxxx" from other sections.
# Section [Include/xxxx] can be located before or after sections that point to it.
#
# Sections are processed from top to bottom, thus the order is important.
# Multiple sections with the same name are allowed, they are merged together.
# Sections may contain the same key multiple times; the last entry wins.
#
# [Default] should be a catch-all action and come last.
#
1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 17: Просмотр файла расширений

The image shows a terminal window of the Midnight Commander (MC) file manager. The title bar indicates the user is 'aaharlamova' and the current directory is '~/newdir'. The main window displays the 'File' menu, which lists various actions like 'Do something on the current file', 'Do something on the tagged files', 'Copy', 'Move', etc. The menu is presented in a text-based format with a light blue background and black text. A mouse cursor is visible over the menu options. The bottom of the window shows a status bar with keyboard shortcuts for various actions: 1 Help, 2 Save, 3 Mark, 4 Replac, 5 Copy, 6 Move, 7 Search, 8 Delete, 9 PullDn, 10 Quit.

```
mc [aaharlamova@aaharlamova]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.m14V18
aaharlamova@aaharlamova:~/work/study/2025-2026/Операцио mc [aaharlamova@aaharlamova]:~/newdir — /usr/bin/mc -P /
menu [----] 0 L: [ 1+ 0 1/380] *(0 /12204b) 0115 0x073 [*][X]
shell_patterns=B
#####
# %% The % character
# %f The current file (if non-local vfs, file will be copied locally and
# %p The current file
# %d The current working directory
# %s "Selected files"; the tagged files if any, otherwise the current file
# %t Tagged files
# %u Tagged files (and they are untagged on return from expand_format)
# %view Runs the commands and pipes standard output to the view command
# If %view is immediately followed by '(', recognize keywords
# ascii, hex, nroff and unform
#
# If the format letter is in uppercase, it refers to the other panel
#
# With a number followed the % character you can turn quoting on (default)
# and off. For example:
# %f quote expanded macro
# %lf ditto
# %0f don't quote expanded macro
#####
+ ! t t
@ Do something on the current file
  CMD=(Enter command)
  $CMD %f
+ t t
@ Do something on the tagged files
  CMD=(Enter command)
  for i in %t ; do
1 Help 2 Save 3 Mark 4 Replac 5 Copy 6 Move 7 Search 8 Delete 9 PullDn 10 Quit
```

Рисунок 18: Просмотр файла меню

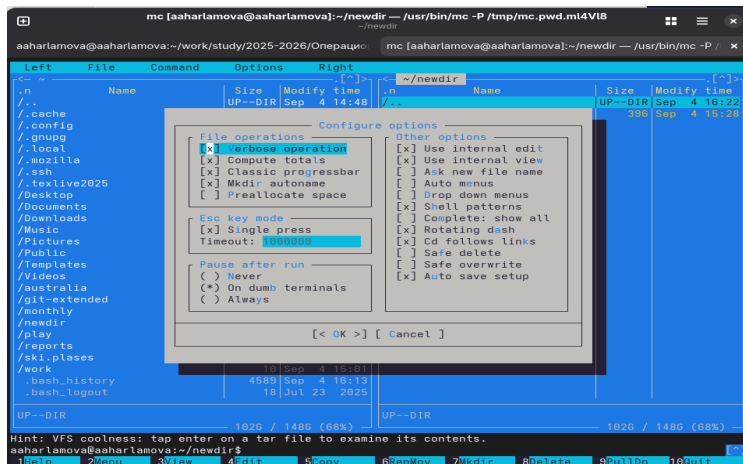


Рисунок 19: Конфигурация

Работа с Midnight Commander

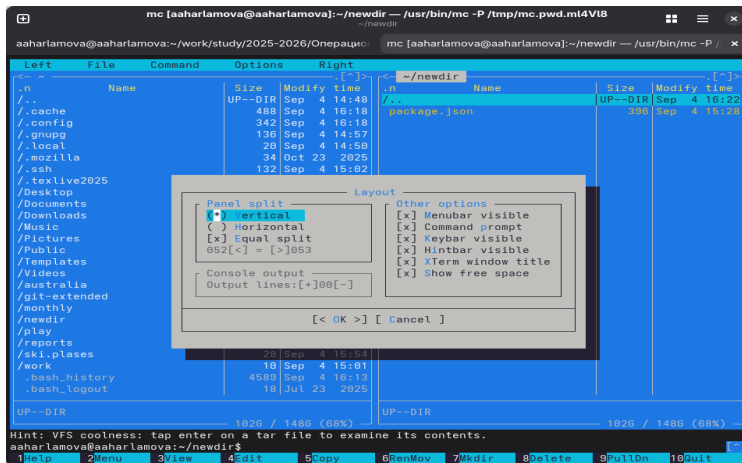


Рисунок 20: Внешний вид

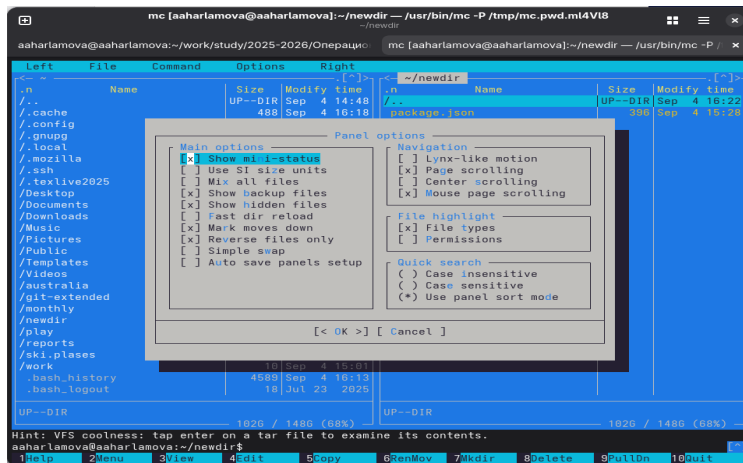


Рисунок 21: Настройки панелей

Работа с Midnight Commander

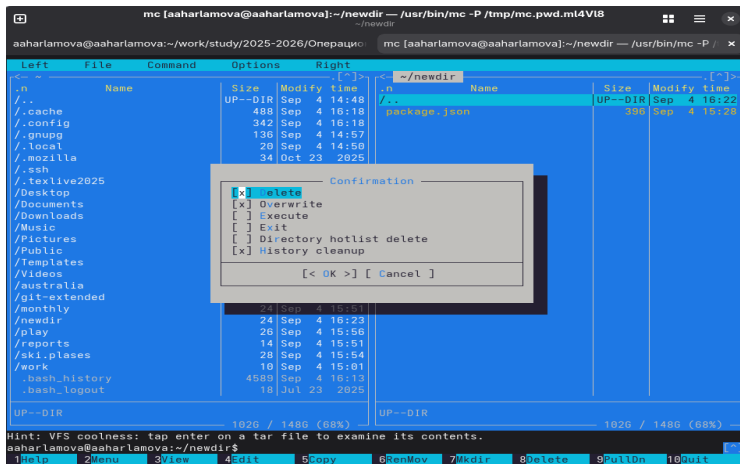


Рисунок 22: Подтверждение

Работа с Midnight Commander

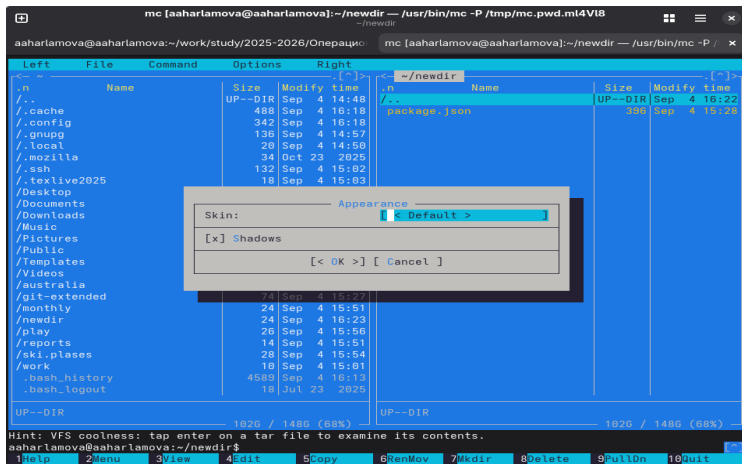


Рисунок 23: Оформление

Работа с Midnight Commander

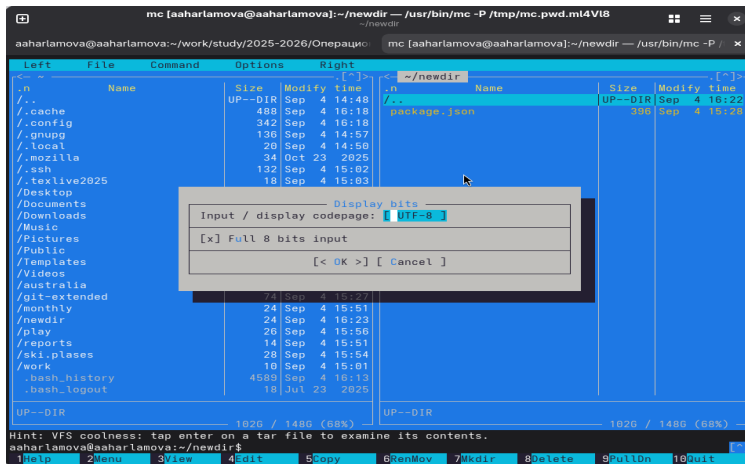


Рисунок 24: Кодировка символов

Работа с Midnight Commander

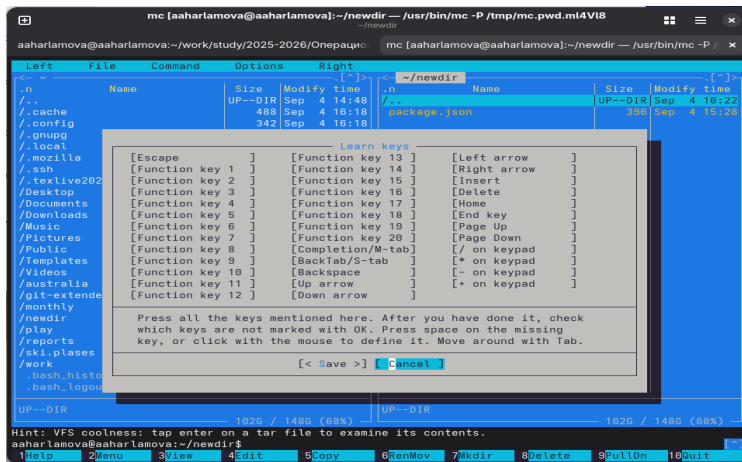
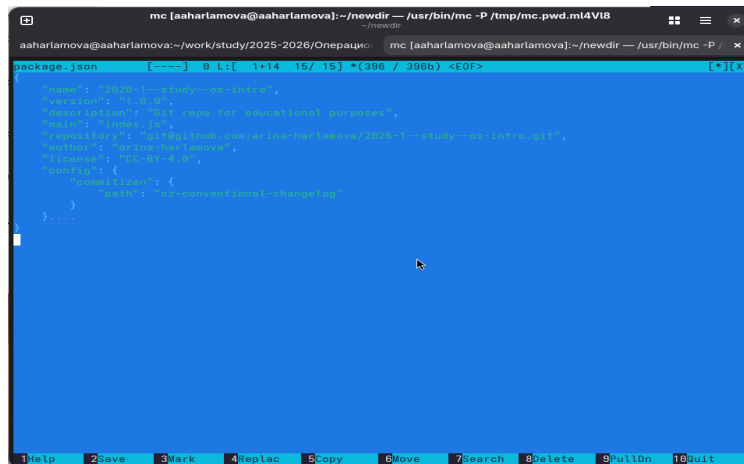


Рисунок 25: Распознавание клавиш

Работа с редактором Midnight Commander



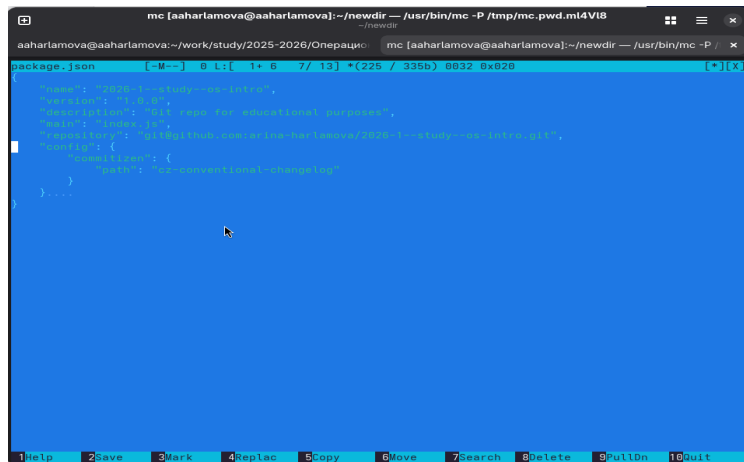
The screenshot shows the Midnight Commander (MC) file editor interface. The title bar indicates the current directory is `~/newdir` and the file being edited is `package.json`. The editor displays the content of `package.json`, which is a JSON file for a project named "2026-1--study--cs-intro". The file content is as follows:

```
package.json [----] 8 L: [ 1+14 15/ 15] *(396 / 396b) <EOF> [ + ][ X ]
{
  "name": "2026-1--study--cs-intro",
  "version": "1.0.0",
  "description": "Bit repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:arina-harlamova/2026-1--study--cs-intro.git",
  "author": "arina-harlamova",
  "license": "CC-BY-4.0",
  "config": {
    "commitizen": {
      "path": "cz-conventional-changelog"
    }
  }
}
```

At the bottom of the window, there is a menu bar with the following options: 1Help, 2Save, 3Mark, 4Replac, 5Copy, 6Move, 7Search, 8Delete, 9PullDn, 10Quit.

Рисунок 26: Файл с текстом

Работа с редактором Midnight Commander



The screenshot shows the Midnight Commander (MC) interface. The title bar at the top reads "mc [aaharlamova@aaharlamova]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.m4Vl8". Below the title bar, the current directory is shown as "aaharlamova@aaharlamova:~/work/study/2025-2026/Операцио". The main window displays the contents of a file named "package.json". The file content is as follows:

```
package.json  [-M--]  0 L: [ 1+ 6 7/ 13] *(225 / 335b) 0032 0x020  [*](X)
{
  "name": "2025-1--study--os-intro",
  "version": "1.0.0",
  "description": "git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:aaharlamova/2025-1--study--os-intro.git",
  "config": {
    "conventional": {
      "path": "conventional-changelog"
    }
  },
  "scripts": {
  },
  "devDependencies": {
  },
  "keywords": [
  ],
  "author": "Aaharlamova",
  "license": "MIT"
}
```

At the bottom of the window, there is a menu bar with the following options: 1 Help, 2 Save, 3 Mark, 4 Replac, 5 Copy, 6 Move, 7 Search, 8 Delete, 9 PullDn, 10 Quit.

Рисунок 27: Файл с текстом

Работа с редактором Midnight Commander

The screenshot shows a terminal window with the following details:

- Terminal Title Bar:** mc [aaharlamova@aaharlamova]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.ml4Vl8
- Terminal Path Bar:** aaharlamova@aaharlamova:~/work/study/2025-2026/Операционные системы → mc [aaharlamova@aaharlamova]:~/newdir — /usr/bin/mc -P /
- File Name:** package.json
- File Size:** 0 L: [1+ 4 5/ 21] *(121 / 751b) 0032 0x020
- File Content:**

```
{
  "name": "2026-1--study--os-intro",
  "version": "1.0.0",
  "description": "Bit repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:arina-harlamova/2026-1--study--os-intro.git",
  "main": "index.js",
  "repository": "git@github.com:arina-harlamova/2026-1--study--os-intro.git",
  "main": "index.js",
  "repository": "git@github.com:arina-harlamova/2026-1--study--os-intro.git",
  "main": "index.js",
  "repository": "git@github.com:arina-harlamova/2026-1--study--os-intro.git",
  "main": "index.js",
  "repository": "git@github.com:arina-harlamova/2026-1--study--os-intro.git",
  "main": "index.js",
  "repository": "git@github.com:arina-harlamova/2026-1--study--os-intro.git",
  "config": {
    "conventional": {
      "path": "os-conventional-changelog"
    }
  },
  ....
}
```
- Terminal Footer:** 1 Help 2 Save 3 Mark 4 Replac 5 Copy 6 Move 7 Search 8 Delete 9 PullDn 10 Quit

Рисунок 28: Копирование фрагмента

Работа с редактором Midnight Commander

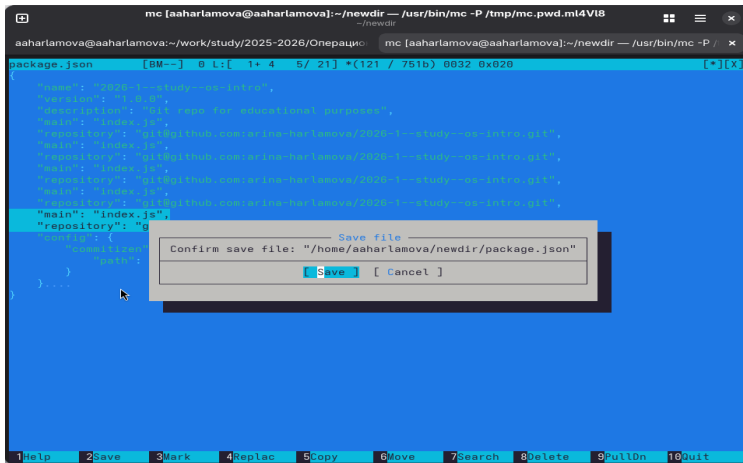
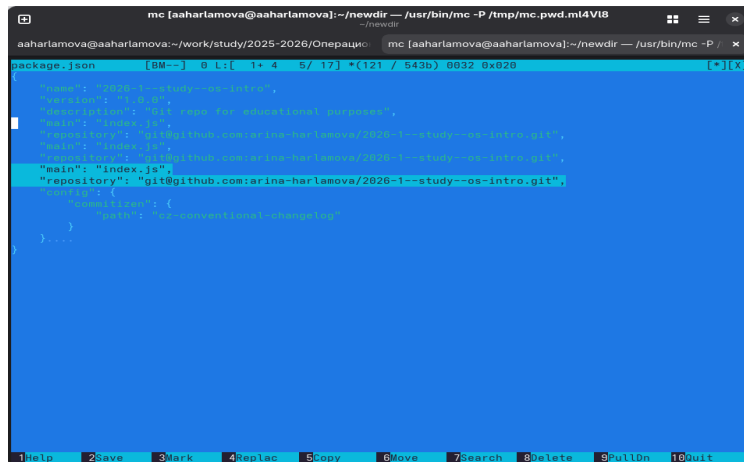


Рисунок 29: Сохранение

Работа с редактором Midnight Commander



```
mc [aaharlamova@aaharlamova]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.ml4Vl8
aaharlamova@aaharlamova:~/work/study/2025-2026/Операцио: mc [aaharlamova@aaharlamova]:~/newdir — /usr/bin/mc -P /
package.json [BM--] 0 L:[ 1+ 4 5/ 17] +(121 / 543b) 0032 0x020 [*][X]
{
  "name": "2026-1--study--os-intro",
  "version": "1.0.0",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:arina-harlamova/2026-1--study--os-intro.git",
  "main": "index.js",
  "repository": "git@github.com:arina-harlamova/2026-1--study--os-intro.git",
  "main": "index.js",
  "repository": "git@github.com:arina-harlamova/2026-1--study--os-intro.git",
  "config": {
    "convention": {
      "path": "cz-conventional-changelog"
    }
  }
}
```

Рисунок 30: Отмена

Работа с редактором Midnight Commander

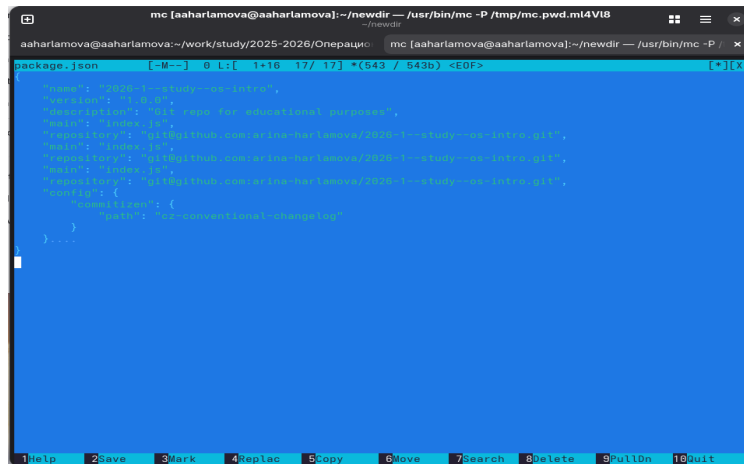


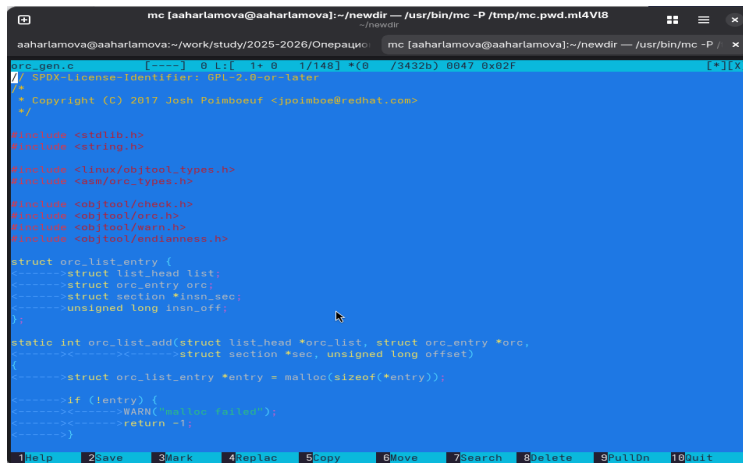
Рисунок 31: Переход в конец файла

Работа с редактором Midnight Commander

```
mc [aaharlamova@aaharlamova]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.ml4Vl8
~ /newdir
aaharlamova@aaharlamova:~/work/study/2025-2026/Операцио mc [aaharlamova@aaharlamova]:~/newdir — /usr/bin/mc -P / ×
package.json [-M--] 1 L: [ 1+ 0 1/ 17] *(1 / 543b) 0010 0x00A [*][X]
{
  "name": "2025-1--study--os-intro",
  "version": "1.0.0",
  "description": "Bit repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:arina-harlamova/2025-1--study--os-intro.git",
  "main": "index.js",
  "repository": "git@github.com:arina-harlamova/2025-1--study--os-intro.git",
  "main": "index.js",
  "repository": "git@github.com:arina-harlamova/2025-1--study--os-intro.git",
  "config": {
    "conventional": {
      "path": "cz-conventional-changelog"
    }
  },
  "scripts": {
    "test": "echo \"Error: no test specified\" & exit 1"
  }
}
1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 32: Переход в начало файла

Работа с редактором Midnight Commander



```
mc [aaharlamova@aaaharlamova]:~/newdir — /usr/bin/mc -P /tmp/mc.pwd.mt4Vl8
aaharlamova@aaaharlamova:~/work/study/2025-2026/Операцио: mc [aaharlamova@aaaharlamova]:~/newdir — /usr/bin/mc -P /

orc_gen.c  [----]  8 L: [ 1+ 8 1/148] +(8 /3432b) 8047 0x02F  [*][X]
/*
 * SPDX-License-Identifier: GPL-2.0-or-later
 *
 * Copyright (C) 2017 Josh Poimboeuf <jpoimboe@redhat.com>
 */

#include <stdlib.h>
#include <string.h>

#include <linux/objtool_types.h>
#include <asm/orc_types.h>

#include <objtool/check.h>
#include <objtool/orc.h>
#include <objtool/warn.h>
#include <objtool/endianness.h>

struct orc_list_entry {
    struct list_head list;
    struct orc_entry orc;
    struct section *insn_sec;
    unsigned long insn_off;
};

static int orc_list_add(struct list_head *orc_list, struct orc_entry *orc,
    struct section *sec, unsigned long offset)
{
    struct orc_list_entry *entry = malloc(sizeof(*entry));

    if (!entry) {
        WARN("malloc failed");
        return -1;
    }
}
```

Рисунок 33: Файл с программой

Работа с редактором Midnight Commander

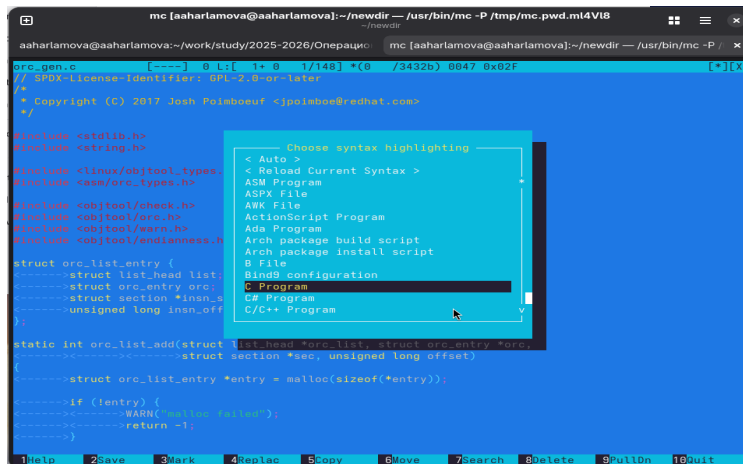


Рисунок 34: Цветовыделение синтаксиса

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.